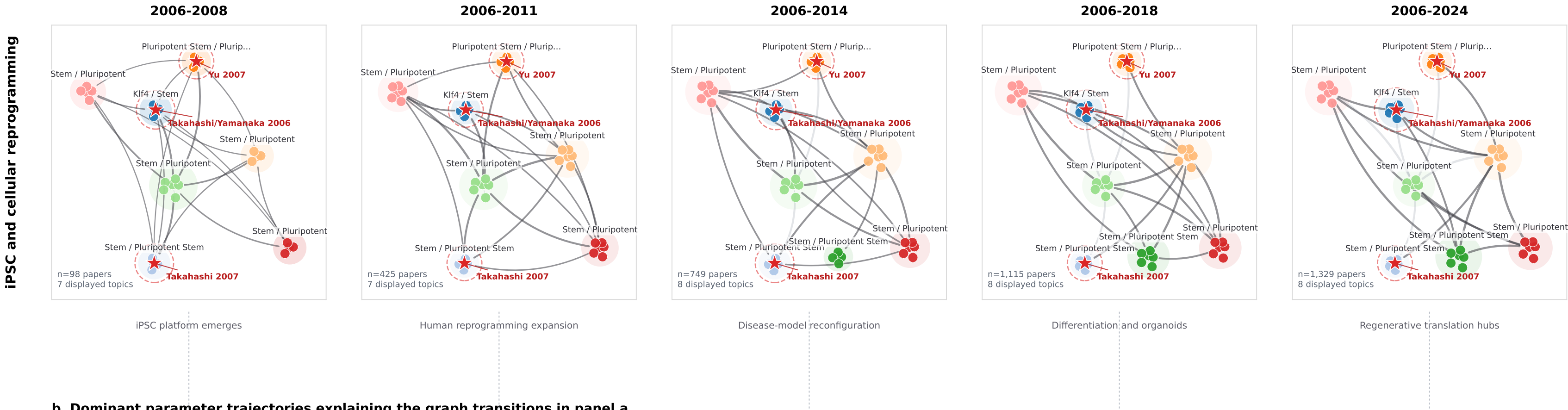


Fig. 1 | Citation knowledge-graph perturbation after a landmark innovation

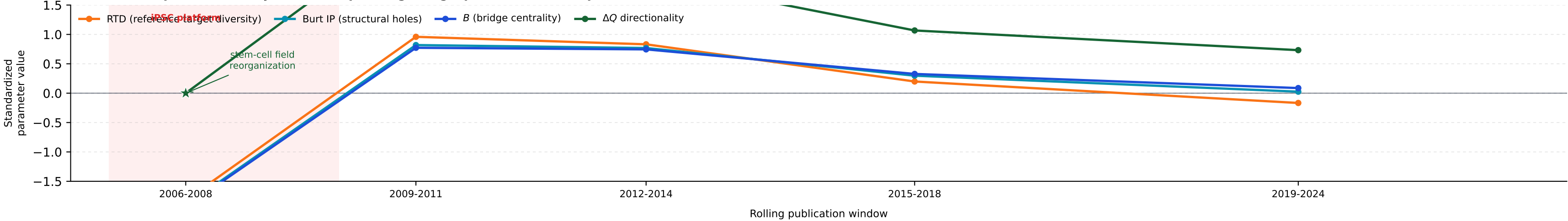
Example domain: iPSC and cellular reprogramming. Real OpenAlex data; publication-style display uses data-derived topic snapshots.



a Cumulative citation knowledge graph snapshots



b Dominant parameter trajectories explaining the graph transitions in panel a



Recombination
RTD up + Burt IP up
Structural holes are recombined

Bridging
 B up
Somatic identity meets pluripotency

Reorganization
 $\Delta Q < 0$
Stem-cell boundaries are reorganized